

Homework 2 — The Language of Sets

Week 2

Name: _____

Instructions

Use **pen**. Write questions and surprises in the margin in a different color. Some things in the reading are intentionally imprecise — find them. This reading is difficult, if there are topics you don't understand, you should use the internet or AI to help. AI can be a good teacher if you ask it specific questions. **DO NOT** use AI to solve the problems without trying yourself first.

1 Readings

Take notes in your notebook on important definitions and examples that you find helpful. Taking notes is a skill in itself. As you read the following content, ask questions in your head (using what you learned from the **question compass**) and try answering it. If you can't answer your own question, re-read the text. If you still can't answer it after re-reading, write your question down in a **different color pen** in the margins and we will go over in it class.

Maximal Interpretation

Every Clue Earns Its Place

In class we looked at a problem about two Minecraft bedrooms. One student understood the problem so that Bedroom 2's width (8 blocks) was also the length of the shared wall — a perfectly valid logical interpretation. But in that reading, the last sentence of the problem ("you haven't decided how wide Bedroom 1 will be yet") was doing nothing. No variable was needed. No unknown existed.

That interpretation wasn't *wrong* — it just wasn't **maximal**.

Definition. An interpretation is **maximal** if every piece of given information is used. An interpretation that leaves a clue unused is called **wasteful**.

In a well-written problem, every sentence is there for a reason. Before you commit to a reading, ask yourself: *Is every piece of given information doing something in my interpretation?* If a sentence is just sitting there, look for a better reading.

There is a hierarchy of how badly an interpretation can fail:

Type	What it means
Contradiction	Your interpretation makes two clues logically impossible at the same time. Ruled out entirely.
Wasteful	Your interpretation is consistent, but leaves at least one clue unused. Probably wrong — keep looking.
Maximal	Every clue is active. This is the interpretation to work with.

Note: A maximal interpretation is not always unique. If two different readings each use all the given information, both deserve consideration — and the problem itself may be ambiguous. That is worth flagging too. Often times it is just a poorly written question.

Interpretation Practice

I1. Write a problem or statement such that there exists a maximal interpretation and wasteful interpretation (can be more than two). Explain the possible interpretations and what makes them maximal versus wasteful.

What Is a Set?

Definition. A **set** is an unordered collection of **distinct** objects. Each object is called an **element** (or **member**) of the set.

$a \in A$ means “ a is an element of A ”

$a \notin A$ means “ a is NOT an element of A ”

Example. Let $S = \{1, 3, 5\}$. Then $3 \in S$, but $4 \notin S$.

One important fact: a set has **no repeated elements** and **no order**. The sets $\{1, 2, 3\}$, $\{3, 1, 2\}$, and $\{1, 1, 2, 3\}$ are all the same set.

Two Ways to Write a Set

Roster notation — list the elements explicitly:

$$A = \{2, 4, 6, 8\}$$

Set-builder notation — describe the property elements must satisfy:

$$A = \{x \in \mathbb{Z} \mid x \text{ is even and } 0 < x \leq 8\}$$

Read the vertical bar $|$ as “**such that.**” The full phrase: “*the set of all x in $\mathbb{Z}$ such that x is even and between 0 and 8.*”

Example. $\{x \in \mathbb{R} \mid x^2 = 9\} = \{-3, 3\}$. Why are there two elements? Because both $(-3)^2 = 9$ and $3^2 = 9$.

The Number Sets You Will See Constantly

Symbol	Name	Description
$\mathbb{N}$	Natural numbers	$\{1, 2, 3, 4, \dots\}$
$\mathbb{Z}$	Integers	$\{\dots, -2, -1, 0, 1, 2, \dots\}$
$\mathbb{Q}$	Rational numbers	All fractions $\frac{p}{q}$ with $p, q \in \mathbb{Z}$, $q \neq 0$
$\mathbb{R}$	Real numbers	Every point on the number line
$\emptyset$	The empty set	$\{ \}$ — a perfectly valid set containing nothing

These sets are **nested**: $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$. Every natural number is an integer, every integer is rational, and so on.

[Number-classification diagram omitted from the public release.]

Subsets

Set A is a **subset** of set B , written $A \subseteq B$, if every element of A is also in B .

Formally: $A \subseteq B$ means: for all x , if $x \in A$ then $x \in B$.

Examples:

- $\{2, 4\} \subseteq \{1, 2, 3, 4, 5\}$? **Yes.** Both 2 and 4 appear.
- $\{2, 7\} \subseteq \{1, 2, 3, 4, 5\}$? **No.** $7 \notin \{1, 2, 3, 4, 5\}$.
- $\{1, 2, 3\} \subseteq \{1, 2, 3\}$? **Yes.** they are equal in this case.
- $\emptyset \subseteq \{1, 2, 3\}$? **Yes.** All sets have nothing in it.

Set Operations

Given sets A and B :

Operation	Notation	Meaning
Union	$A \cup B$	Everything in A OR B (or both)
Intersection	$A \cap B$	Everything in BOTH A AND B
Difference	$A \setminus B$	In A but NOT in B
Complement	A^c	Everything in U that is NOT in A

Definition. A **universal set** U is a declared set that contains every object under consideration. All other sets in the problem satisfy $A \subseteq U$.

Example. If we are working with integers from 1 to 10, we write $U = \{1, 2, 3, \dots, 10\}$. Any set we discuss — say $A = \{2, 4, 6\}$ — must be a subset of U .

Why it matters. Union, intersection, and set difference depend only on A and B . But complement depends on U :

$$A^c = \{x \in U \mid x \notin A\}$$

Change U , and A^c changes. Without declaring U , the complement is undefined.

Example. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$.

$$A \cup B = \{1, 2, 3, 4\}, \quad A \cap B = \{2, 3\}, \quad A \setminus B = \{1\}, \quad A^c = \{4, 5\}$$

Definition. The **cardinality** of a set A , written $|A|$, is the number of elements in A .

Using the example above, fill in each cardinality:

$ A $	$ B $	$ A \cup B $	$ A \cap B $

Cartesian Products

Definition. The **Cartesian product** $A \times B$ (read “ A cross B ”) is the set of ALL ordered pairs (a, b) where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example. $A = \{1, 2\}$, $B = \{\text{red, blue, green}\}$:

$$A \times B = \{(1, \text{red}), (1, \text{blue}), (1, \text{green}), (2, \text{red}), (2, \text{blue}), (2, \text{green})\}$$

So $|A \times B| = 2 \times 3 = 6$.

You already know one Cartesian product: the coordinate plane is $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. Every point (x, y) is an ordered pair — one from each copy of $\mathbb{R}$.

Big preview. In the next worksheet, we define a *function* $f: A \rightarrow B$ as a special subset of $A \times B$ — a set of ordered pairs where each $a \in A$ appears exactly once on the left. Keep that in mind. By the end of next class, this sentence will make complete sense.

Why You Cannot Just Add

Suppose 13 students like Math and 9 students like Science. Can you conclude that 22 students like at least one subject?

Not necessarily — some students might like *both*. If 5 students like both, then adding $13+9 = 22$ counts those 5 students twice: once in the Math group and once in the Science group. To fix the overcount, subtract the overlap once. In this situation, there are $13 + 9 - 5 = 17$ students that like at least one subject. Notice mathematically what we did was:

$$|\text{Math} \cup \text{Science}| = |\text{Math}| + |\text{Science}| - |\text{Math} \cap \text{Science}| = 13 + 9 - 5 = 17$$

Also notice that the phrase “students that like **at least** one subject” translates to

$$|\text{Math} \cup \text{Science}|.$$

The Inclusion-Exclusion Formula

The previous problem demonstrates an important formula in set theory, the **inclusion-exclusion formula**. This works for any two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Why it works. Every element of $A \cap B$ gets added in $|A|$, added again in $|B|$, then subtracted once. Net result: counted exactly once. Every element in only A or only B is never subtracted. Net result: also counted exactly once.

Example. On a Minecraft server, 13 players have built a house and 9 have defeated the Ender Dragon. 5 players have done both. How many players have done at least one?

$$|H \cup D| = 13 + 9 - 5 = 17$$

If there are 25 players total, then $25 - 17 = 8$ have done neither.

Practice. At a summer camp, 24 campers went hiking and 19 went swimming. 8 campers did both. There are 40 campers total.

(a) How many campers did at least one activity?

(b) How many campers did neither?

Foundation Practice

1. Membership, Notation, and Set-Builder Form.

(a) Let $S = \{3, 6, 9, 12, 15\}$. Fill in $\in$ or $\notin$:

6 ____ S	0 ____ S	9 ____ S	18 ____ S
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(b) Write S from (a) in set-builder notation using $\mathbb{N}$. (Hint: what property do ALL elements of S share?)

(c) Write each in roster notation.

(i) $\{x \in \mathbb{Z} \mid -2 \leq x \leq 2\}$ = _____

(ii) $\{x \in \mathbb{Z} \mid x^2 < 10\}$ = _____

(iii) $\{x \in \mathbb{N} \mid x \text{ divides } 12\}$ = _____

(d) Write $\{x \in \mathbb{R} \mid x^2 = 4\}$ in roster notation.

$$\{x \in \mathbb{R} \mid x^2 = 4\} = \underline{\hspace{2cm}}$$

2. Subsets.

(a) List ALL subsets of $\{p, q, r\}$. Include the empty set.

How many subsets are there total? _____

If you see a pattern, try to state it in terms of the number of elements in the original set.

(b) Answer T / F and justify in one sentence.

T / F $\emptyset \subseteq \{1, 2, 3\}$

T / F $\{1, 2, 3\} \subseteq \{1, 2, 3\}$

T / F $\{1, 2\} \subseteq \{1, \{2\}\}$ *(Look carefully at the second set.)*

(c) Fill in the blanks, ordering $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ from “smallest” to “largest” using $\subseteq$:

_____ $\subseteq$ _____ $\subseteq$ _____ $\subseteq$ _____

Circle all that apply for each number:

Number	$\in \mathbb{N}$?	$\in \mathbb{Z}$?	$\in \mathbb{Q}$?
-5	Yes No	Yes No	Yes No
$\frac{1}{3}$	Yes No	Yes No	Yes No
$\sqrt{2}$	Yes No	Yes No	Yes No
0	Yes No	Yes No	Yes No

3. Cardinality.

(a) Compute the cardinality of each set.

$ \{a, e, i, o, u\} $	$ \emptyset $	$ \{\emptyset\} $

Note: $\emptyset$ is a set with zero elements. $\{\emptyset\}$ is a set that **contains** the empty set — it has exactly one element. Are they the same set? Explain.

(b) Suppose $|A| = 4$ and $|B| = 6$. Use $|A \cup B| = |A| + |B| - |A \cap B|$.

(i) What is the **maximum** possible $|A \cup B|$? When does this happen?

(ii) What is the **minimum** possible $|A \cup B|$? When does this happen?

(c) ★ *Preview.* If $|A| = m$ and $|B| = n$, give a formula for $|A \times B|$. Explain why the formula makes sense — don't just guess.

Part 2 — Application

4. Set Operations.

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$.

(a) Complete the table.

Expression	Result (roster notation)
$A \cup B$	
$A \cap B$	
$A \setminus B$	
$B \setminus A$	
A^c (in U)	
$(A \cup B)^c$ (in U)	

(b) Is $A \setminus B$ the same as $B \setminus A$? Answer and explain.

(c) Verify the formula $|A \cup B| = |A| + |B| - |A \cap B|$ using your table. Show the arithmetic.

(d) The reading defined complement as “everything NOT in A .” But question 4(a) says “ A^c in U ” — why did we need to specify U ? What would A^c mean if we used $U' = \mathbb{N}$ as the universal set instead?

5. Real Life Application — Minecraft Achievement Board.

You manage the achievement board for a Minecraft server with 30 active players. After a big weekend event, you collect the following data:

- 18 players have built a house. Call this set H .
- 14 players have defeated the Ender Dragon. Call this set D .
- 6 players have done **both**.

(a) How many players earned at least one achievement?

(b) How many players earned **neither** achievement?

(c) Write an expression in set notation for “players who built a house but did NOT defeat the Ender Dragon,” then compute the count.

_____ = _____ players

(e) A third achievement is added: Nether Portal (set P) with $|P| = 10$. The server rules require having a house before you can build a portal, so $P \subseteq H$.

(i) Can you simplify $P \cup H$? What is it equal to? Explain.

(ii) What do you know about $P \cap H$? Can you simplify it?

(iii) What is the minimum possible value of $|P \cap D|$? The maximum? (You do not have information about P and D specifically — think about what's possible.)

Part 3 — Challenge

6. Cartesian Products.

Let $A = \{0, 1, 2\}$ and $B = \{\text{red}, \text{blue}\}$.

(a) List all elements of $A \times B$.

(b) List all elements of $B \times A$.

(c) Is $A \times B = B \times A$? If not, give a specific element that is in one but not the other.

(d) The coordinate plane is $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. Every point (x, y) on the grid is an element of $\mathbb{R}^2$. Write the x -axis as a *subset* of $\mathbb{R}^2$ in set-builder notation.

$x\text{-axis} = \{ \rule{10cm}{0.4pt} \}$

(e) *Preview.* Consider the function $f: \{1, 2, 3\} \rightarrow \{x, y, z\}$ defined by $f(1) = x$, $f(2) = x$, $f(3) = z$. Write f as a subset of $\{1, 2, 3\} \times \{x, y, z\}$ — just list the ordered pairs.

$$f = \{ \rule{10cm}{0.4pt} \}$$

7. Your First Set Theory Proof.

We are going to prove that for any sets A and B , $A \cap B \subseteq A$.

(a) Fill in the blanks to complete the proof.

Proof.

Let $x \in A \cap B$. *(We introduce an arbitrary element of $A \cap B$.)*

By the definition of intersection, $x \in \rule{1.5cm}{0.4pt}$ and $x \in \rule{1.5cm}{0.4pt}$.

Therefore, $x \in \rule{1.5cm}{0.4pt}$. *(This is what we needed.)*

Since x was arbitrary, $A \cap B \subseteq A$. Q.E.D.

(b) Write the analogous proof that $A \cap B \subseteq B$. (Same template — do it yourself from scratch.)

(c) Is $A \subseteq A \cap B$ always true? Give a **proof** if yes, or a **counterexample** if no. A counterexample means you find specific sets A and B that break it.

8. What does it mean for something to be **countable**? Provide a formal mathematical **definition** on what it means to be countable?

Part 4 — Notebook Artifact

N1. For the notes you took while doing the readings/homework (I hope you did), write at least 5 questions about any topic today in a different color pen in your notebook.

N2. If you used the internet or AI for help, describe how you used it:

ANSWER KEY — DO NOT DISTRIBUTE

Worksheet I-B: The Language of Sets

Grading notes: credit full marks for correct mathematical reasoning even if notation differs from key. Require justification for T/F and proof questions.

Reading — In-line Answers

I1 (Interpretation Practice). Open-ended. Accept any problem with a clear maximal interpretation and at least one wasteful alternative. A strong response names the specific clue that goes unused in the wasteful reading and explains why the maximal reading requires it.

Cardinality fill-in (reading table). From the example $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$: $|A| = 3$, $|B| = 3$, $|A \cup B| = 4$, $|A \cap B| = 2$.

Inclusion-Exclusion Practice (summer camp). U = all 40 campers at the summer camp.

(a) $|H \cup S| = 24 + 19 - 8 = \mathbf{35}$ campers.

(b) $40 - 35 = \mathbf{5}$ campers did neither.

Venn diagram regions: hiking only = 16, both = 8, swimming only = 11, neither = 5. Check: $16 + 8 + 11 + 5 = 40$. ✓

Reading Planted Imprecision

Complement without U : The operations table lists complement as “everything NOT in A ” without mentioning U . The complement is only well-defined relative to a universal set. Q4(d) forces students to confront this.

$\mathbb{N}$ includes 0? The reading states $\mathbb{N} = \{1, 2, 3, \dots\}$ (no 0). Some textbooks define $\mathbb{N}$ to include 0 (especially in logic and computer science). This is a legitimate ambiguity. Neither definition is universally “correct.” Good class discussion: what happens to the $\mathbb{N} \subseteq \mathbb{Z}$ chain if 0 is in $\mathbb{N}$? (Nothing breaks — $0 \in \mathbb{Z}$ either way.)

Foundation

Q1.

(a) $6 \in S$, $0 \notin S$, $9 \in S$, $18 \notin S$

(b) $S = \{x \in \mathbb{N} \mid x \text{ is a multiple of 3 and } x \leq 15\}$

(c)

(i) $\{-2, -1, 0, 1, 2\}$

(ii) $\{-3, -2, -1, 0, 1, 2, 3\}$ $((-3)^2 = 9 < 10$, so ± 3 qualify)

(iii) $\{1, 2, 3, 4, 6, 12\}$

(d) **Curveball.** $x^2 = 4$ has two solutions: $x = 2$ and $x = -2$. Answer: $\{-2, 2\}$. Students who write only $\{2\}$ forgot negative solutions.

Q2.

(a) $\emptyset$, $\{p\}$, $\{q\}$, $\{r\}$, $\{p, q\}$, $\{p, r\}$, $\{q, r\}$, $\{p, q, r\}$

Total: **8** subsets. Pattern: a set with n elements has 2^n subsets. *Why 2^n : for each element, independently choose include or exclude — 2 choices per element, n elements, so 2^n total.*

(b)

T / F $\emptyset \subseteq \{1, 2, 3\}$: **TRUE**. The empty set is a subset of every set — there are no elements in $\emptyset$ that fail to be in $\{1, 2, 3\}$. (Vacuously true.)

T / F $\{1, 2, 3\} \subseteq \{1, 2, 3\}$: **TRUE**. Every set is a subset of itself.

T / F $\{1, 2\} \subseteq \{1, \{2\}\}$: **FALSE**. *Curveball.* The set $\{1, \{2\}\}$ has two elements: the number 1 and the *set* $\{2\}$. The number 2 is not an element of $\{1, \{2\}\}$.

(c) $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$

Table: -5 : No, Yes, Yes. $\frac{1}{3}$: No, No, Yes. $\sqrt{2}$: No, No, No (irrational). 0 : No (per our definition), Yes, Yes.

Q3.

(a) $|\{a, e, i, o, u\}| = 5$; $|\emptyset| = 0$; $|\{\emptyset\}| = 1$.

$\{\emptyset\}$ contains exactly one element: the empty set itself. $\emptyset \neq \{\emptyset\}$. *Analogy: an empty bag and a bag containing an empty bag are different objects.*

(b)

(i) Maximum $|A \cup B| = 10$. Happens when $A \cap B = \emptyset$ (disjoint).

- (ii) Minimum $|A \cup B| = 6$. Happens when $A \subseteq B$.
- (c) $|A \times B| = m \cdot n$. For each of the m elements of A , pair with each of the n elements of B : n pairs per element, m elements total $\Rightarrow m \cdot n$.

Part 2 — Application

Q4.

Expression	Answer
$A \cup B$	$\{1, 2, 3, 4, 5, 6, 8\}$
$A \cap B$	$\{2, 4, 6\}$
(a) $A \setminus B$	$\{1, 3, 5\}$
$B \setminus A$	$\{8\}$
A^c (in U)	$\{7, 8, 9, 10\}$
$(A \cup B)^c$ (in U)	$\{9, 10\}$

- (b) No. $A \setminus B = \{1, 3, 5\}$, $B \setminus A = \{8\}$. Set difference is not commutative.
- (c) $|A \cup B| = 7$. Formula: $6 + 4 - 3 = 7$. ✓
- (d) **Planted imprecision.** If $U = \mathbb{N}$, then $A^c = \{7, 8, 9, 10, 11, 12, \dots\}$ — an infinite set. The complement changes completely depending on U .

Q5.

- (a) $|H \cup D| = 18 + 14 - 6 = \mathbf{26}$ players.
- (b) $30 - 26 = \mathbf{4}$ players.
- (c) $H \setminus D = 18 - 6 = \mathbf{12}$ players.
- (d)
- (i) $P \cup H = H$. Since $P \subseteq H$, every element of P is already in H .
- (ii) $P \cap H = P$. Since $P \subseteq H$, every element of P is in both.
- (iii) Minimum $|P \cap D| = \max(0, 10 + 14 - 30) = 0$. Maximum $|P \cap D| = \min(10, 14) = 10$.

So $0 \leq |P \cap D| \leq 10$.

Part 3 — Challenge

Q6.

(a) $A \times B = \{(0, \text{red}), (0, \text{blue}), (1, \text{red}), (1, \text{blue}), (2, \text{red}), (2, \text{blue})\}$

(b) $B \times A = \{(\text{red}, 0), (\text{red}, 1), (\text{red}, 2), (\text{blue}, 0), (\text{blue}, 1), (\text{blue}, 2)\}$

(c) No. $(0, \text{red}) \in A \times B$ but $(0, \text{red}) \notin B \times A$.

(d) $x\text{-axis} = \{(x, 0) \mid x \in \mathbb{R}\}$

(e) $f = \{(1, x), (2, x), (3, z)\}$. y is NOT in the range.

Q7.

(a) Blanks: A and B ; therefore A .

(b) Let $x \in A \cap B$. By the definition of intersection, $x \in A$ and $x \in B$. Therefore, $x \in B$.
 $\square$

(c) **FALSE.** Counterexample: $A = \{1, 2\}$, $B = \{2, 3\}$. $A \cap B = \{2\}$, but $1 \in A$ and $1 \notin A \cap B$. So $A \not\subseteq A \cap B$.

Note: $A \subseteq A \cap B \iff A \subseteq B$.

Q8. A set S is **countable** if there exists an injective function $f: S \rightarrow \mathbb{N}$ — the elements of S can be listed without repetition. Finite sets are countable. $\mathbb{Z}$ and $\mathbb{Q}$ are countable. $\mathbb{R}$ is **not** countable (Cantor's diagonal argument — see historian sheet).